Jing-Zhong Wang

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Research Interests

Measuring population health and its inequalities: Bayesian hierarchical and spatiotemporal models for small-area mortality and life expectancy, cause-specific forecasting, decomposition of health gaps, and burden of disease; scalable Bayesian computation (INLA, PyMC/NUTS on GPU).

Education

National Taiwan University — Taipei, Taiwan

M.S. in Biostatistics and Health Data Science — *Sep 2022 – Jun 2024*

B.S. in Public Health — *Sep 2018 – Jun 2022*

- **M.S. thesis:** Life Expectancy Disparities at Township Level in Taiwan: Bayesian Spatio-Temporal Analysis. Supervisor: Prof. Hsien-Ho Lin.
- **Certificate programmes:** Biological Statistics; Big Data in Health; Infectious Diseases.
- **Selected coursework:** Measuring Burden of Disease, Methods in Descriptive Epidemiology, Spatial Analysis, Modelling of Infectious Disease Epidemiology, Repeated Measurements, Categorical Data Analysis, Applied Linear Algebra, Machine Learning.

Publications and Manuscripts

Wang J-Z in bold. †Co-senior authors.

Published

[1] Fan S-P, Liu P-C, **Wang J-Z**, Win NN, Lin C-H, Lin H-H. Epidemiological burden of Alzheimer's and Parkinson's diseases in East Asia: a comparative review of global burden of disease estimates and national registry data. *Journal of the Formosan Medical Association.* 2026;125:S5–S12. doi:10.1016/j.jfma.2026.03.072

[2] Fan S-P, Liu P-C, **Wang J-Z**, Lin C-H, Lin H-H. Disease burden of neurodegenerative disorders in Taiwan. *Journal of the Formosan Medical Association.* 2026;125:S105–S115. doi:10.1016/j.jfma.2026.01.020

Under review

[3] **Wang J-Z**, Liu P-C, Bennett JE, Lo W-C, Lu T-H, Ezzati M, Chiang T-L, Lin H-H. Evolving drivers of geographic inequalities in life expectancy under universal health coverage: a nationwide spatiotemporal analysis of Taiwan, 2000–2021. *European Journal of Public Health* (Revision submitted after major revision, Sep 2026).

[4] **Wang J-Z**, Nomura S, Chang AY†, Lin H-H†. Uneven progress in survival across high-income Asia: a comparative analysis of age, cause, and geographic inequalities. (Submitted, Oct 2026).

[5] Chen C-H, Liu P-C, **Wang J-Z**, Lu F-W, Yang W-K, Hill AM, Lo W-C, Jeng J-S, Lin H-H. National trends in stroke burden in Taiwan from 2008–2023: incidence, mortality, and disability-adjusted life years. (Submitted, Jul 2026).

Research Experience

Institute of Epidemiology and Preventive Medicine, National Taiwan University — Taipei, Taiwan

Research Associate · PI: Prof. Hsien-Ho Lin — *Mar 2021 – Oct 2024; Feb 2025 – Present*

Nov 2024 – Jan 2025: mandatory military service, Taiwan.

- **Global mortality projection (main analyst).** Built an end-to-end pipeline projecting mortality for 27 cause groups in 186 countries from WHO Global Health Estimates (2000–2023) to 2035: nine INLA age–period–cohort models combined by Bayesian stacking, regional borrowing for data-sparse countries, probabilistic life tables, and a Global Health 2050 scenario with 15 priority conditions.
- **Subnational life expectancy, Taiwan (first author).** Built township-level Bayesian spatiotemporal models (358 townships, 2000–2021; ICAR + RW effects, PyMC/NUTS on GPU) to show how drivers of geographic inequality evolved under universal health coverage; co-authored with Prof. Majid Ezzati (Imperial College London).
- **Comparative life expectancy, high-income Asia (first author).** Decomposed survival gains in Japan, Korea, Singapore and Taiwan (2000–2019) by age, cause and region, and estimated subnational life expectancy in Japan, Korea and Taiwan. Co-senior authors: Prof. Angela Y. Chang (University of Southern Denmark) and Prof. Hsien-Ho Lin.
- **Multimorbidity and burden of disease** (NTU Core Consortium project, 2025–2027). Used association rule mining to compare observed and expected co-occurrence of 29 GBD conditions in national insurance claims for all adults (2006–2023) by year, sex and age, with FDR control; showed that the independence assumption in burden studies overstates comorbidity-adjusted YLDs of these conditions by about 3%.
- ***H. pylori* and gastric cancer burden.** Synthesised 1,700+ studies in a Bayesian meta-analysis of *H. pylori* prevalence and estimated attributable and non-attributable gastric cancer burden with subnational INLA models.

School of Public Health, Imperial College London London, UK
Short-term research visit (research meetings) *Mar 2025 – Apr 2025*

- Funded through Prof. Hsien-Ho Lin's research project. Held research meetings with Prof. Majid Ezzati and Dr James Bennett on Bayesian small-area mortality modelling for the township-level life expectancy study of Taiwan.

Department of Biostatistics, Harvard T.H. Chan School of Public Health Remote
Research Intern *Feb 2021 – Oct 2021*

- NTU public health field practicum. Project: *On aspects of high-dimensional mediation analysis with applications to epigenetic studies* (practicum poster, 2021). Identified mediators in high-dimensional epigenetic data using penalised regression (MCP, Lasso) linking toxicant exposure to gestational age; applied causal-inference frameworks. Supervisors: Prof. Rajarshi Mukherjee (Harvard) and Prof. Wan-Chen Lee (NTU).

Government-Commissioned Research Projects

National Health Insurance Administration, Ministry of Health and Welfare 2025 – Present
Estimating the burden of non-communicable disease in Taiwan (PI: Wei-Cheng Lo) *Research Associate*

- Pre-processed NHI claims for case ascertainment and post-processed burden estimates (prevalence, YLLs, YLDs, DALYs) for 2010–2023.
- Designed and built the project's interactive visualisation platform (R Shiny, highcharter): treemaps of YLL, YLD and DALY by disease hierarchy, time trends for 2010–2023, age patterns and subnational maps, with filters by year, sex, age and region.

Department of Health, Taipei City Government 2024
Disease and mortality profile of Taipei residents (PI: Wei-Cheng Lo; co-PI: Hsien-Ho Lin) *Part-time RA*

- District-level NCD burden (prevalence, mortality, DALYs) for Taipei, including comparisons between Indigenous and non-Indigenous residents, using NHI claims and death registration data.

Health Promotion Administration, Ministry of Health and Welfare Aug 2022 – Jul 2023
Analytical models and standardised methods for NCD burden estimation (PI: Hsien-Ho Lin) *Part-time RA*

- Analysed NHIRD claims and cause-of-death registry data to estimate diabetes prevalence, YLLs, YLDs and DALYs by year, age, sex and township, following the GBD framework.
- Wrote SAS programs, run on site at the Health and Welfare Data Science Center, that identify chronic kidney disease in NHIRD claims (dialysis, transplant, ICD-9/10 diagnoses, erythropoietin use) and export point-prevalent cases by stage, year (2000–2021), sex, age and county for YLD estimation.
- Estimated population attributable fractions of diabetes for high BMI and physical inactivity (comparative risk assessment) and reviewed international use of burden estimates in health policy for the expert panel.

Software and Projects

- **letools** *(R)*: R package for life tables (Kannisto, Gompertz and constant-hazard closeouts) and Arriaga, Pollard and cause-specific decomposition, with Chiang confidence intervals and ∼100 unit tests; consolidates code reused across projects.
- **bayesianHPmodel** *(R, INLA)*: Reproducible workflow for subnational Bayesian models of *H. pylori* prevalence and gastric cancer burden.
- **LEvis** *(JavaScript, Highcharts)*: Interactive maps of life expectancy in 358 Taiwanese townships, 2000–2021 (live demo, simulated data).
- **Global mortality projection dashboard** *(R Shiny, highcharter)*: Country-level mortality and life-expectancy projections and scenarios for 186 countries, with choropleth maps and cached data.

Teaching

- **Instructor**, Taiwan Burden of Disease Center workshops, NTU: Introduction to R (Dec 2021); Processing risk-factor survey data in R (Dec 2023); Data visualisation in R (Sep 2024); Advanced data visualisation (Jul 2025).
- **Teaching assistant**, NTU: Biostatistics I (3 semesters); Applied Biostatistics (1 semester).

Honours and Awards

- Excellence Award, Research Poster Exhibition, Institute of Epidemiology and Preventive Medicine, NTU (M.S. thesis) 2024
- 3rd Place, English Oral Presentation (M.S. thesis research) 2023

Skills

Methods	Bayesian hierarchical and spatiotemporal models (ICAR/BYM2, RW, AR, APC), Bayesian stacking, meta-analysis, life tables and demographic decomposition, disease burden (PAF, DALY), causal inference
Computing	R (INLA, data.table, tidyverse, ggplot2, Shiny), Python (PyMC, nutpie, JAX on GPU, scikit-learn), SAS, SQL; Git, Linux/HPC batch workflows
Data	Taiwan NHIRD and vital registration, WHO Global Health Estimates, GBD
Languages	Mandarin (native), English (professional), German (intermediate), Japanese (beginner)